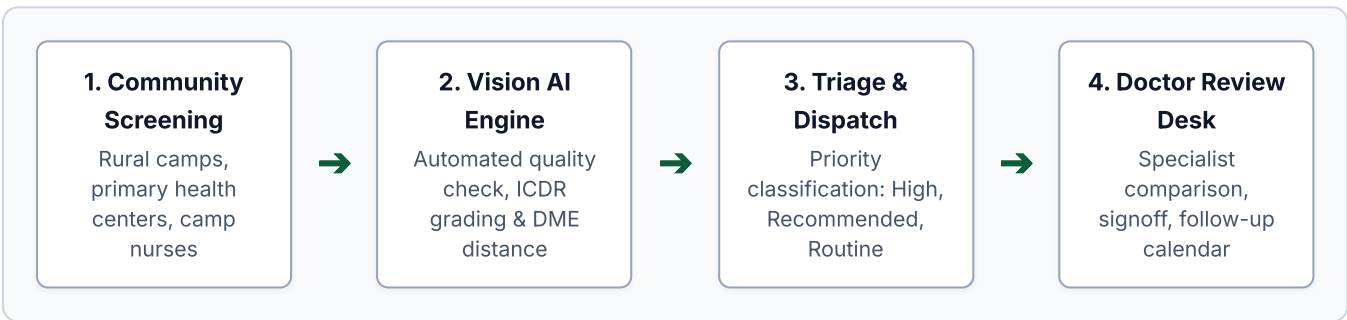


Clinical Core Tenet: "AI assists. Doctors decide."

RetinoAI is built with strict role separation. Frontline healthcare workers screen with real-time photographic quality checks and diagnostic guidance, while hospital-based retina specialists maintain complete diagnostic and clinical referral authority.

1. System Architecture & Portal Separation

RetinoAI operates as a dual-portal teleophthalmology platform that bridges remote screening camps with urban tertiary eye hospitals:



Portal A: Clinical Screening Portal (Frontline Operators)

7-Stage Guided Workflow: (1) Patient Demographics Registry → (2) Diabetes Biomarkers (HbA1c %, duration, medication) → (3) Visual Acuity & Ocular Symptoms (IOP, blurriness, floaters) → (4) Patient Consent & Data Release → (5) Multi-Point Retinal Photography & Quality Score → (6) AI Vision Screening Results → (7) Specialist Escalation.

Portal B: Doctor Review Portal (Ophthalmologists & Retina Specialists)

Specialist Review Desk: Automated triage queue sorted by clinical risk; high-magnification fundus canvas with crosshair coordinates; **Longitudinal Visit Comparison** placing historical screening visits side-by-side with today's visit; one-click referral order (*Refer to Tertiary Hospital vs Routine Annual Monitor*) and follow-up calendar provisioning.

2. Medical Vision & Clinical Diagnostic Engine

The vision engine evaluates high-resolution fundus photographs using international standards established by the American Academy of Ophthalmology (AAO) and the Early Treatment Diabetic Retinopathy Study (ETDRS):

A. International Clinical Diabetic Retinopathy (ICDR) 5-Stage Disease Scale

GRADE	CLINICAL STAGE	PATHOLOGICAL FINDINGS ON FUNDOSCOPY	RECOMMENDED ACTION
0	No DR	Normal caliber retinal vessels, intact macula, clear optic nerve margins, healthy foveal reflex.	Annual routine re-screening
1	Mild NPDR	Microaneurysms only (isolated focal capillary out-pouchings along temporal arcades).	Repeat screening in 6–12 months
2	Moderate NPDR	More than microaneurysms: scattered blot hemorrhages in 1–2 quadrants, early hard lipid exudates.	Comprehensive dilated fundoscopy
3	Severe NPDR	4-2-1 Rule: Severe blot hemorrhages across all 4 quadrants, venous beading in ≥ 2 quadrants, or prominent IRMA in ≥ 1 quadrant.	Urgent specialist referral (within 1–2 weeks)
4	PDR	Active neovascularization of disc (NVD) or elsewhere (NVE), fibrous proliferation, preretinal/vitreous hemorrhage risk.	Urgent surgical / anti-VEGF consult (within 48–72 hrs)

B. ETDRS Macular Edema (CSME) Classification

Diabetic Macular Edema (DME) is classified independently from the vascular retinopathy stage:

- **None:** No retinal thickening or hard exudates in posterior pole.
- **Mild:** Exudates or retinal thickening distant from the macular center.
- **Clinically Significant Macular Edema (CSME):** Triggered when hard exudates or retinal thickening occur **within 500 µm** (1/3 disc diameter) of the central fovea, or when circinate lipid plaques envelope the foveal avascular zone.

C. Green-Channel Optical Calibration (540–570 nm)

Oxygenated and deoxygenated hemoglobin have an absorption peak at 540–570 nm (green wavelength spectrum). The RetinoAI engine applies green-channel contrast enhancement (1.48:1 contrast ratio) to demarcate microvascular leaks, punctate aneurysms, and subtle micro-hemorrhages from the darker choroidal background.

3. Multi-Model AI Gateway & IDE Integration

RetinoAI utilizes a resilient, multi-provider artificial intelligence architecture:

PROVIDER / ENGINE	MECHANISM & MODEL	OPERATIONAL MODE
Calibrated Clinical Engine	Deterministic local vision heuristics & biomarker scoring	Default (100% Offline): Requires zero API keys or external network connection. Zero latency.
Google Gemini Vision	Gemini 2.0 Flash / 1.5 Flash multimodal vision API	Free Cloud AI: Activated via <code>GEMINI_API_KEY</code> . Analyzes live fundus photos with high-resolution visual tokens.
Blackbox AI Gateway	Blackbox enterprise vision endpoint (<code>blackbox-pro</code>)	Cloud AI: Activated via <code>BLACKBOX_API_KEY</code> . Supports visual lesion detection and code copilot queries.
OpenAI-Compatible IDE Bridge	<code>POST /api/chat/completions</code> & <code>/v1/chat/completions</code>	IDE Integration: Connects Cline, Blackbox Agent, or Continue extensions directly to RetinoAI on port 5000.

4. Technology Stack & Verification Results

The solution is built as a production-grade TypeScript monorepo with strict linting, zero build errors, and an automated integration test suite:

- **Frontend Portal:** React 19, TypeScript, Vite 6, Swiss-Editorial UI, interactive SVG/Canvas fundus viewer.
- **Backend Server:** Node.js 20+, Express 5, TypeScript strict mode, pino-http logging.

- **Atomic Persistence:** Thread-safe file swapping (`.tmp` → rename replacement) in `PersistentStore` to guarantee zero data corruption under concurrent operations.

Automated Integration Test Results (12/12 Passing — 100% Pass Rate)

- ✓ GET `/api/healthz` (Healthcheck) → PASS
- ✓ POST `/api/patients` (Register Patient) → PASS
- ✓ GET `/api/patients/:id` (Retrieve Patient) → PASS
- ✓ POST `/api/screening-cases` (Intake & Biomarkers) → PASS
- ✓ POST `/api/screening-cases/:id/images` (Upload Fundus OD/OS) → PASS
- ✓ POST `/api/screening-cases/:id/ai-analysis` (ICDR Staging & DME) → PASS
- ✓ POST `/api/screening-cases/:id/escalate` (Specialist Referral) → PASS
- ✓ GET `/api/referred-cases` (Doctor Worklist Queue) → PASS
- ✓ POST `/api/screening-cases/:id/review` (Specialist Signoff) → PASS
- ✓ GET `/api/follow-ups` (Follow-up Registry) → PASS
- ✓ GET `/api/datasets` (Benchmark Datasets Registry) → PASS
- ✓ POST `/api/datasets/import-case` (Ingest Calibrated Case) → PASS

Clinical Safety Notice: RetinoAI is an assistive diagnostic decision-support system. All AI findings, lesion annotations, and priority triage designations require review and signoff by a licensed medical practitioner before therapeutic intervention.